

Piezoelectric Actuation Systems: Optimization of Driving Electronics

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ABSTRACT

Voltage control is by far the most common strategy for driving piezoelectric actuators, but it is, at times, handicapped by large, inefficient amplifiers and nonlinear piezoelectric actuator behavior. Optimizing voltage-feedback amplifiers for driving capacitive-type loads with amplifier schemes such as pulse-width-modulation promises to drastically reduce amplifier weight and improve efficiency. This development work is currently in progress and is discussed in this paper.

Also presented are some methods that take advantage of direct charge control of piezoelectric actuators. Direct charge control removes much of the hysteresis that is inherent in voltage control so it is a likely option if a high level of positioning accuracy is needed in a given application. Unlike voltage-feedback control, charge amplifiers can accurately control a piezoactuator open loop. The main drawback to charge control is that periodic removal of charge bias is necessary to prevent actuator drift. Discussed are strategies that attempt to accomplish charge control while simultaneously avoiding actuator drift and conserving power.

Keywords: piezoelectric, power, dissipation, efficiency, amplifier, electronics, smart structures, smart materials, control

1. INTRODUCTION

Piezoelectrics have often been touted as the solution to actuation problems in smart structures applications because of their small size, high power density and their ability to act as both a sensor and actuator. However, what is often neglected is the fact that the electronics that control the piezoelectric can often be bulky and inefficient. The strategies that these components use to control the deflection and/or force output of the piezoactuator fall into two main categories, voltage control and charge control. Improving the performance and efficiency of the electronics at the heart of both control strategies is the focus of this research effort.

2. VOLTAGE CONTROL STRATEGIES

Piezoelectric actuators represent a particularly difficult electric load. A first order approximation often used models the piezoelectric as a linear capacitor. In fact, piezoelectric actuators represent much more complex loads in that they contain numerous resonant (and anti-resonant) frequencies, and they generate a voltage in response to a stress or strain. The circuit shown in Figure 1 is an electric circuit equivalent for a piezoelectric, as given by the IEEE Standards on Piezoelectricity [1]. In this circuit C_o is the static or parallel capacitance and the parallel RLC branch represents one electromechanical resonance. In the small region near a particular resonance, R_i , L_i , and C_i are constant. In order to model other resonances additional RLC branches can be added. By adding additional RLC components, a fairly accurate model of the piezoelectric is generated. For the majority of power electronic design situations, the piezoelectric load is assumed to be purely capacitive. In this case the static capacitance (and voltage level) determines the power requirements of the electronics. If a piezoelectric is to be excited at a resonance frequency then one RLC branch can be added to increase the accuracy of the model. Adding more RLC branches is typically not necessary.

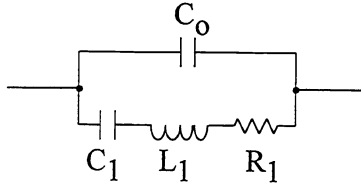


Figure 1. Equivalent circuit of a piezoelectric near a resonance.

The most commonly used method of controlling piezoelectrics is through an applied voltage. The critical assumption made in voltage control of piezoelectric actuators is that the electric field is the applied voltage divided by the piezoelectric thickness, $E=V/t$. This assumption yields the relationship

$$S = s^E T + d \frac{V_{out}}{t} \quad (1)$$

where S is strain, s^E is elastic compliance at constant total field, T is stress, d is a piezoelectric constant, V_{out} is the applied voltage and t is the wafer thickness.

This is a very convenient equation because it gives the output strain, S , in terms of the stress on the piezomaterial, T , and the voltage applied to the electrodes, V_{out} . Since the displacement output of a piezoelectric actuator is proportional to the strain, S , Eq. (1) is called the piezoelectric actuator equation. Traditionally this equation is used as the governing equation for piezoelectric actuator.

One issue with voltage control of piezoelectrics that arises is predicting the correspondence between the amplifier input and the actuator displacement output. Equation (1) implies that piezoactuators can be driven open loop—that the output displacement will be a linear function of the input voltage. In the real world, however, this is not the case. The behavior of piezoelectric actuators under voltage control can be extremely hysteretic, necessitating some sort of position feedback if positioning to a high level of accuracy is desired. This hysteresis arises from the fact that an assumption was made equating the field in the piezoelectric material to the field *applied* to the piezoelectric material ($E=V_{out}/t$) [2].

While the operation of a voltage controlled piezoelectric actuator is not one-to-one, it is inherently stable. Therefore, direct voltage control of piezoelectric elements is perfectly feasible if the inherent hysteresis is within tolerable limits.

2.1 Analog Voltage Control

The most commonly used strategy for control of piezoelectric actuators is analog voltage control; that is, driving the piezoelectric actuator with an amplifier that uses a simple applied voltage-feedback strategy. A simplified schematic of this control circuit is shown in Figure 2 with the amplifier represented in a basic inverting configuration. In this case the voltage applied to the actuator will be related to the voltage applied to the amplifier input by the relationship

$$V_{out} = -\frac{R_f}{R_i} V_{in} \quad (2),$$

where V_{out} and V_{in} are the output and input voltages of the amplifier, respectively, and R_f and R_i are the values of the feedback and input resistors.

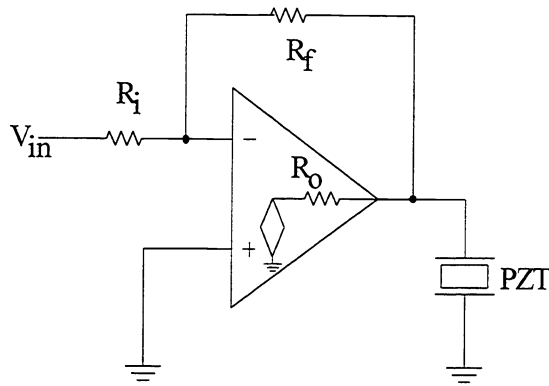


Figure 2. A typical voltage-feedback inverting power amplifier.

With minor variations, the schematic in Figure 2 represents the vast majority of amplifiers that are currently used in piezoelectric systems. From a laboratory testing point of view these amplifiers are the easiest to use because they are readily available in the form of benchtop power amplifiers. Analysis of the circuit in Figure 2 reveals some significant problems with this approach, however.

In order to estimate the power requirements of the amplifier, the piezoelectric actuator is assumed to be a pure capacitive load, C . The capacitance used for amplifier design is the parallel or static capacitance of the piezoelectric, which can be readily measured and is a function of the size, shape, and material properties of the piezoelectric. The specified displacement and/or force output of the piezoelectric actuator determines the required voltage, V . The current requirement is given by,

$$I = 2\pi f_{max} VC \quad (3),$$

where f_{max} is the maximum driving frequency. The current requirement given by this equation is considered to be the worst case situation and an amplifier is typically chosen accordingly.

When a piezoelectric actuator is being driven, large currents are flowing in and out of the amplifier output, and thus through the output resistor, R_o . In fact, most of the power in the system is not consumed as actuator work, but is dissipated in R_o where it does no useful work at all. When driving a capacitive load, the current and voltage are 90° out of phase. This phase difference results in significant power being dissipated within the amplifier. The power dissipation requirements of the power amplifier can be estimated by

$$PD_{out,max} = \frac{V_s^2}{2Z_L} \left| \frac{4}{\pi} - \cos \theta \right| \quad (4)$$

where V_s is the amplifier supply voltage, Z_L is the magnitude of the load impedance, and θ is the phase angle of the load impedance. For a purely capacitive load Eq. (4) becomes

$$PD_{out,max} = 4V_s^2 f_{max} C \quad (5)$$

Large current requirements and dissipation of large amounts of power in the output resistor has the effect of adding weight in the form of heat sinks on the amplifier.

The power amplifier requirements can be illustrated by considering a specific piezoelectric stack actuator. A piezoelectric stack actuator used for optical positioning on the order of $20 \mu\text{m}$ has a capacitance of $1 \mu\text{F}$ and requires 150 V_{p-p} . The optical positioning bandwidth requirement is 2 kHz . From these specifications, the voltage amplifier must supply 950 mA and dissipates 180 W ! This example shows how a seemingly simple application of piezoelectric actuation can become very difficult from an electronics standpoint. The above issues arise partly because traditional power voltage amplifiers are

optimized to drive resistive loads, not piezoelectric actuators. Piezoactuators appear electrically to be more "capacitive" than "resistive".

2.2 Switched Voltage Control

In order to decrease the heat sink requirement in driving non-resonant capacitive loads the switched voltage control strategy can be used. In this method the amplifier output voltage has two states: high and low. An example of a power MOSFET switching circuit is shown in Figure 3(a).

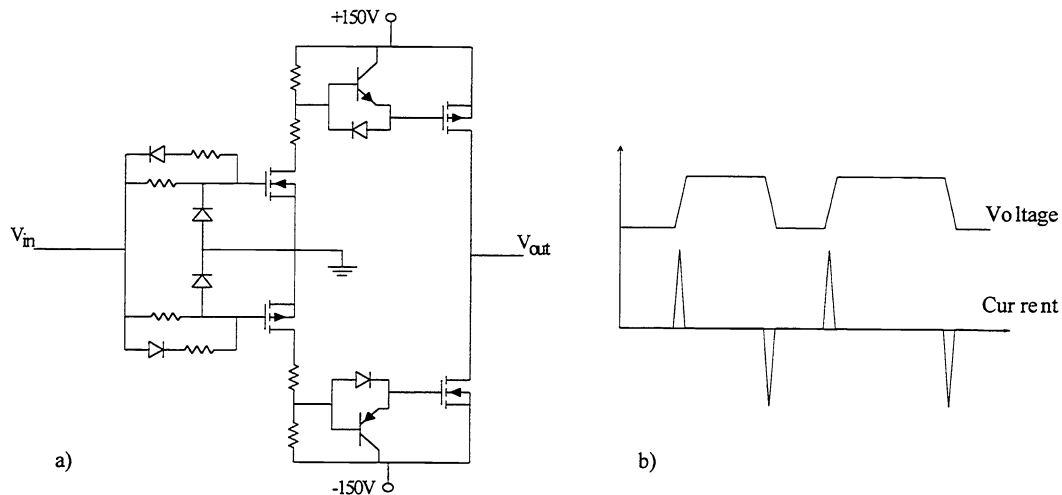


Figure 3. Power MOSFET switch (a) and Voltage and current output (b).

Power savings are realized in the circuit shown above by maintaining a phase angle of 0° or 180° , as shown in Figure 3b. Power is consumed only during the switching from one state to the other. The power savings of a switching amplifier over a continuous voltage amplifier can be as large as two orders of magnitude. In terms of the piezoelectric stack example the power consumption can be reduced to less than 2W. Of course the disadvantage of switched voltage control is that only two output states are available. This problem can be addressed by switching between the two states at a rate high enough that the mechanical system cannot respond. The dwell time of the two respective states will define the actuator displacement. This, of course, is the basic idea behind pulse-width-modulation (PWM) amplifiers. This strategy maintains a 180° phase shift between voltage and current in the power module of the amplifier. The design of a prototype amplifier for piezoelectric applications that uses these PWM principles is shown in Figure 4. Essentially the PWM amplifier has the same power module as the switching amplifier already shown. The power module is driven by a TTL signal which maintains a constant frequency square wave while the duty cycle of the TTL signal is varied. The output of the power module swings from the positive to negative rail according to the TTL signal. This output is fed into a low pass LRC filter in which the piezoelectric actuator acts as the capacitor. The LRC network effectively averages the output of the power module so the voltage across the piezoelectric is proportional to the duty cycle. The averaged voltage output is fed back and compared to the command input to further linearize the PWM amplifier response.

The advantage of this design is the extremely high power output/power dissipated ratio. A conventional linear amplifier capable of driving a load at 100V and 2A could easily require a 200 W heat sink to dissipate the thermal losses. This is in stark contrast to a PWM amplifier, which can do the same job while dissipating only 2-3 W. One critical factor in the design of a PWM amplifier is the switching frequency. A high switching frequency increases the power consumption of the amplifier, while a low switching frequency could excite the piezoelectric system. The gain of the low pass filter at the switching frequency determines the magnitude of the voltage ripple that is seen on the piezoelectric. The PWM amplifier must be designed such that this ripple voltage does not adversely affect the actuation performance.

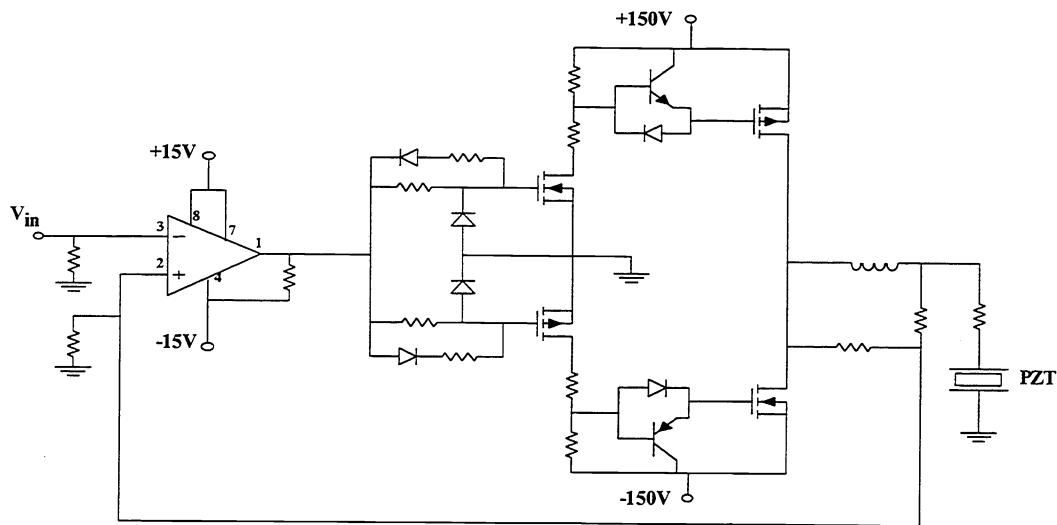


Figure 4. Schematic of a pulse-width modulation voltage-feedback amplifier.

3. CHARGE CONTROL STRATEGIES

All of the voltage-control strategies fail to address one of the fundamental problems with voltage control, the nonlinear relationship between the amplifier input and actuator output. Solving this problem involves completely changing the basic concept of the amplifier designs. It has been shown by a number of researchers that the behavior of a piezoelectric actuator is linear if charge is used as a control quantity rather than voltage [2,3]. The relationship given by Eq. (1) is really only an approximation of the piezoelectric constitutive equations. This approximation represents a fundamental fault with voltage control of piezoelectric actuators and necessitates the incorporation of an additional independent feedback loop if a high degree of absolute positioning is required. One way to avoid hysteresis in the open loop case is to use applied charge as the control variable. The principal fault of charge control as it has been conventionally practiced [3] is that it is fundamentally an integration operation and therefore is susceptible to electronic drift.

The charge-control relationship for piezoelectric materials is developed by eliminating E from the constitutive equations, and developing control equations in terms of the electric displacement. The constitutive equations for a piezoelectric actuator reduce to

$$S_3 = \left(\frac{1}{Y} - \frac{d_{33}^2}{\epsilon} \right) T_3 + \frac{d_{33}Q}{2\epsilon A} \quad (6)$$

under charge control, where Y is Young's modulus, ϵ is the material permittivity, A is the electrode surface area, and Q is the applied charge. Equation (6) is the actuator equation for charge control. In a stress free state the strain is directly proportional to the applied charge.

3.1 Switched Charge Control

A possible alternative to analog charge-feedback control that may abate the drift problem and greatly reduce power requirements is switched charge control. The circuit shown in Figure 5 is composed of four major blocks: the pulse module, the power module, the charge capture module, and the feedback module. All of these sub-circuits, with the exception of the power module can be miniaturized and integrated onto a single IC chip. The basic premise of operation of this circuit is twofold. First, the circuit dynamically delivers (or extracts) charge packets to (from) the piezoelectric load to effect displacement. Second, the circuit monitors and integrates the total charge delivered to the load as a measure of displacement for comparison to a user-defined set point. This comparison generates a feedback error signal to control the desired displacement.

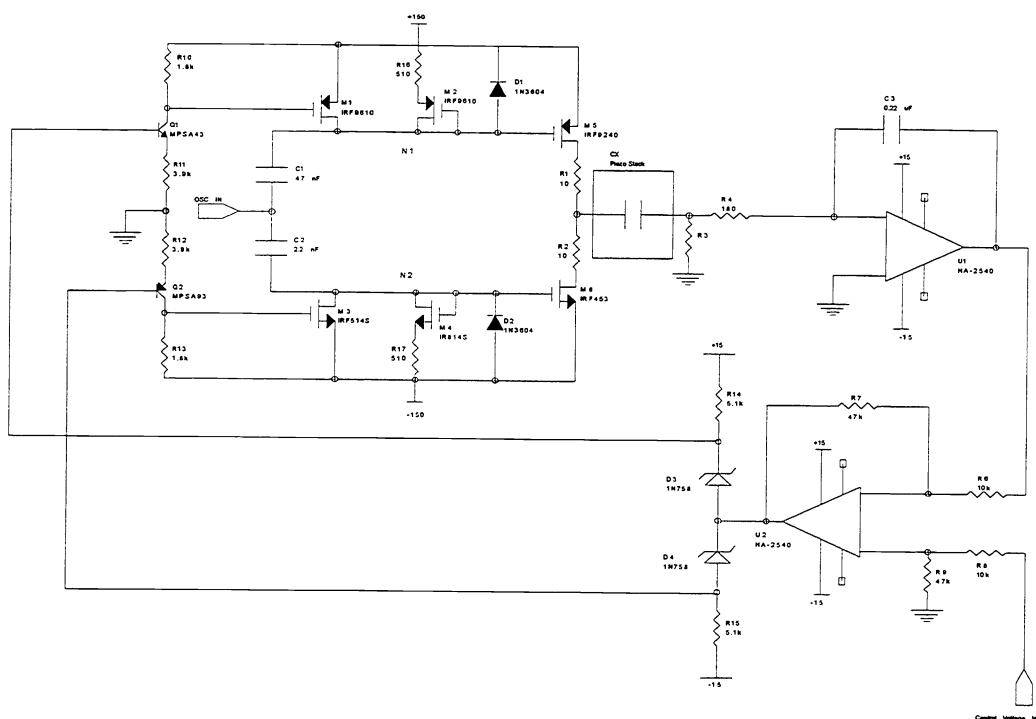


Figure 5. Charge pulse amplifier.

The power module consists of high-voltage, high-current MOSFET devices arranged in a push-pull configuration to supply (source) or extract (sink) charge packets from the piezoelectric load. This charge-based delivery system consumes no static power, only the amount of dynamic power needed to displace the piezoelectric stack is consumed, thus one of the major power consumption issues with the present charge controllers is addressed. Controlling the power module is the pulse module. This sub-circuit uses a feedback error signal (indicating the degree of displacement needed) to generate self-timed voltage pulses to drive the power module. Although the same high voltage power supply is used for this module, the currents are low-power dissipation and related problems (e.g., heat sinking) are negligible. As this is a pulse mode circuit, the pulse module is dynamically controlled by a standard oscillator. The frequency of oscillation controls the delivery timing of the charge packets to the load.

Monitor and control of the position of a piezoelectric actuator is accomplished by the charge capture amplifier and the feedback control circuits. The charge capture amplifier monitors and integrates an attenuated image of the charge delivered to the load in order to generate an output voltage signal proportional to position. In any circuit utilizing charge storage, leakage is inevitable. Periodically a circuit such as this one must be reset so that the charge on the actuator and the integrated charge monitored by the charge capture amp remain in sync. In this circuit this function is performed by MOSFET switches which can accomplish this reset with sufficient speed that the actuator cannot mechanically respond, thus, disturbances of the mechanical portions of the system are avoided. The feedback circuit compares the position voltage of the charge capture amplifier with the user-defined position voltage to generate the error signal. The circuit shown addresses both fundamental difficulties in driving piezoelectric load. Power consumption is reduced by applying a switched amplifier and actuator linearity and hysteresis are addressed through charge control.

4. HYBRID STRATEGIES

A third amplifier design philosophy with respect to piezoelectric systems can be described as a hybrid strategy; that is, it takes advantage of elements of both the voltage and charge control strategies to achieve its ends. The hybrid design presented below is still in the proof-of-principal stage. Experimental data has been obtained using a device prototype that suggests it may provide another low-power alternative to the amplifier designs previously discussed.

The design concept is shown in Figure 6. The amplifier design uses voltage as the control quantity, thus avoiding the potential for integration errors that charge control possesses, but inherent in the design is a recognition that it is the presence of charge that causes piezoactuators to deflect. This fact is the driver behind the presence of the two capacitors in the circuit shown in Figure 6. The operation of this circuit is as follows. At startup, both capacitors (C1 and C2) and the piezoelectric actuator are electrically in parallel with a voltage source. This causes both capacitors and the actuator to charge to a voltage level V . The voltage source is then disconnected. Since the capacitors and the actuator are storage devices they retain their electric charge. Displacements in the actuator can be commanded through the operation of a switch bank that can electrically place the two charged capacitors either in series or parallel with each other, while remaining electrically in parallel with the actuator. Placing the two capacitors in series with each other after the initial charging phase will briefly expose the actuator to a voltage of $2V$, thus causing charge to flow from C2 onto the actuator. This will result in an actuator extension. Switching the capacitor bank back into the parallel configuration will result in the actuator having a higher potential than the capacitor bank, thus charge will flow out of the actuator and be stored on the bank.

The principal advantage of this method is that the same charge is used over and over to displace the actuator. If the value of the capacitors are large relative to the capacitance of the piezoactuator and the actuator is free from stress, the only energy loss during actuator extension and contraction is thermal loss from charge flow through the conductors. A simple manually operated version of this amplifier was breadboarded to test the amplifier concept. The capacitor charging and switching was accomplished with simple SPST mechanical switches and the displacement of the actuator was monitored with a capacitance gage. The capacitors had a value of $100\text{ }\mu\text{f}$ and the actuator was a stack actuator with a measured capacitance of approximately $1\text{ }\mu\text{f}$. Initially all three were charged to a voltage level of 25 volts. This resulted in an initial displacement in the piezoelectric actuator of approximately 3.15 microns.

Hybrid Storage Amplifier Concept

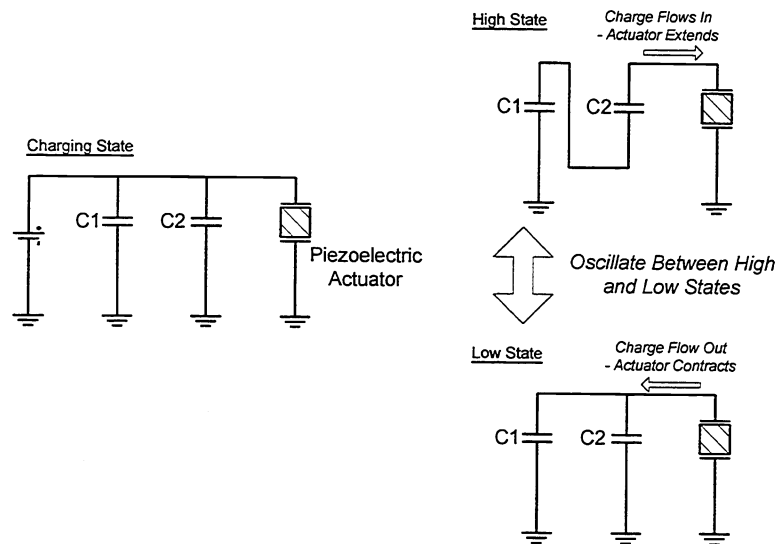


Figure 6. Outline of the Hybrid Storage Amplifier concept.

The results of the switch cycling tests are shown in Figure 7. The actuator was successfully cycled between the high and low states 30 times in this short test. This cycling took place after the capacitors and actuator were charged, but the power supply was no longer connected. The capacitor bank served to salvage the charge used by the actuator during the extension phase and reuse it in subsequent cycles. Although the data supports the concept of charge recovery and storage, the amount of displacement between the high and low states is evidently decreasing as a function of cycle number in this data, indicating that an accommodation will be necessary to inject charge into the circuit as it is slowly dissipated. Future work in this area will concentrate on further development of the amplifier concept, including guidelines on component selection and performance capabilities. Initial indications are that this type of two-state concept may lend itself to a pulse-width-modulation implementation and may provide significant power savings beyond the current state-of-the-art.

Minimum and Maximum Displacement vs. Cycle

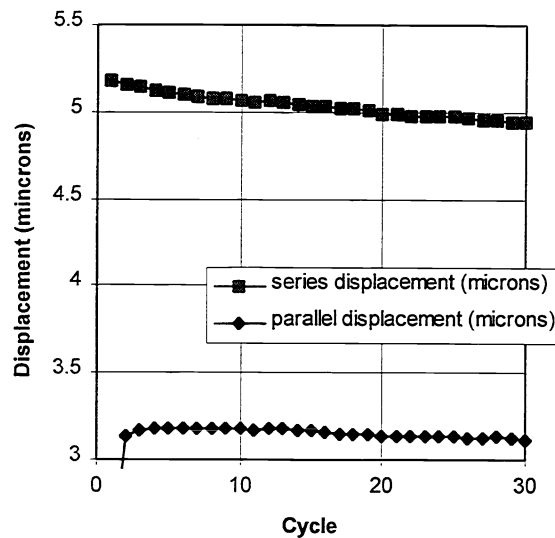


Figure 7. Results of the proof-of-principle Hybrid Storage Amplifier tests.

5. SUMMARY

It is evident that typical benchtop power supplies are badly suited for driving piezoelectric actuator-type loads. The focus of this ongoing investigation is to develop new strategies for power amplifiers that optimize electronics for these loads. Among the strategies discussed the most mainstream at the current time is the voltage control embodiment of the voltage-feedback amplifier. Significant data exists on voltage control PWM's and their behavior in motor control applications. This, combined with the fact that the use of voltage control PWM's can immediately lower the power dissipation in most piezoelectric systems by one or two orders of magnitude, makes them the likely next amplifier-of-choice in piezoelectric applications.

Over the longer term other strategies, such as the pulsed charge or hybrid storage amplifier, may prove to have significant merits. Charge control has been demonstrated as more linear and amplitude-independent than voltage control, but the development of these amplifier designs are still in the prototyping phase.

6 REFERENCES

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